AYDAN J. GERBER

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EDUCATION

Cornell University, Ann S. Bowers College of Computing & Information Science **Ithaca, NY**
Candidate for Bachelor of Science in Information Science; Concentration in Data Science *Expected May 2028*
• SAT: 1570 **GPA:** 3.6/4.0
Relevant Coursework: Object-Oriented Programming and Data Structures, Linear Algebra, Software Design & Development with Python, Statistics and Data Science, User Experience (UX) Design & Prototyping, Finance

PROFESSIONAL EXPERIENCE

SS&C Technologies **Union, NJ**
Software Engineer Intern *June 2026 – August 2026*
- Built an internal LLM assistant answering fund-accounting questions for onboarding staff, citing documents to cut interruptions for senior staff
- Architected a plan-agnostic Python/FastAPI backend abstracting two interchangeable LLM paths behind one streaming endpoint
- Built a self-hosted RAG pipeline (chunking, embeddings, retrieval) returning sourced answers over generic chatbots for grounded answers

REKKIE **Remote**
Algorithm & Data Science Intern *January 2026 – March 2026*
- Built a geospatial ML pipeline converting noisy skier GPS time-series into labeled runs, chairlifts, and waiting states for downstream analytics
- Developed a Hidden Markov Model with Viterbi decoding, engineering speed, heading, slope, and elevation features for route segmentation
- Generated per-run, per-lift, and session analytics (vertical, distance, time, speed) for redesigned iOS and Android app experiences

Outamation **Remote**
Extern *May 2025 – July 2025*
- Built an AI-powered mortgage-document classification system using open-source LLMs (Mistral-7B, Phi-2), automating manual review
- Optimized Retrieval-Augmented Generation pipelines using embeddings, hybrid retrieval, and custom chunking to improve relevance
- Benchmarked model performance with DocLLM and LlamaIndex, delivering recommendations that increased data-extraction accuracy by 30%

theCoderSchool **Scarsdale, NY**
Computer Science Instructor *May 2024 – August 2025*
- Instructed 60 K-12 students in coding through private lessons and group classes, strengthening foundational computer science understanding
- Introduced engineering concepts through LEGO robotics such as gears, motors, pistons, resulting in a deeper understanding of physics
- Fostered coding skills using Java and Python through hands-on interactive coaching and programming concentrating on game development

Iona University **New Rochelle, NY**
Teaching and Lab Assistant, Physics 101 *June 2023 – August 2023*
- Facilitated 12 lab setups, devised a detailed kinematics experiment, and created a comprehensive 20-page instructional manual with diagrams
- Guided 15 students through assignments in kinematics, forces, and energy, resulting in increased understanding of material and concepts
- Repaired 5 pieces of old lab equipment and assisted students throughout lab setup and execution, allowing for better conceptual understanding

RESEARCH EXPERIENCE AND ACTIVITIES

Cornell Data Science Project Team **Ithaca, NY**
Team Member *September 2024 – present*
- Developed LiveDance, a real-time computer vision application integrating pose estimation (MoveNet, BlazePose) with stable skeletal rigging
- Designed a gamified feedback loop with joint scoring, letter grades, and natural-language coaching, turning raw ML output into user guidance
- Collaborated and presented on a hedge-fund–sponsored energy load forecasting project, building BiLSTM forecasters (R^2 0.982, −72% MAE)

Cornell Fintech Club **Ithaca, NY**
Financial Analyst *September 2024 – present*
- Evaluate and present pitches on private tech/SaaS companies (EV under 1B) to invest in and determine appropriate exit strategy for investors
- Collaborated on FinSight AI, a fintech chatbot leveraging Retrieval-Augmented Generation (RAG) to deliver financial information to users
- Drove product iteration through 20+ user interviews validating model outputs and surfacing pain points, improving responses across releases

Cambridge University Press & Assessment **Remote**
Independent Researcher & Author *September 2022 – May 2023*
- Applied topic modeling with Latent Dirichlet Allocation to machine learning study on social media discourse with a focus on sustainability
- Found insights into how consumers and businesses engage with sustainability topics on social media and key trends in discussion on X
- Highlighted the dominant topics of discussion such as global corporate sustainability practices, environmental activism, and green technology
- Paper published in Environmental Data Science journal by Cambridge University Press, DOI: 10.1017/eds.2024.4

AWARDS, SKILLS, & INTERESTS

Awards: Regeneron Science & Engineering Fair (WESEF): 2nd overall & Ricoh Sustainable Development Award, Westchester-Rockland Junior Science & Humanities Symposium 3rd place: Speaker in Behavior III Category

Relevant Skills: Python, Java, Product Strategy, Data Analysis, NLP, User Research, A/B Testing, Communication, Leadership
Interests: Tennis, Weightlifting, Sustainability, Technology, Philosophy, Poker